

GLPI + LibreNMS

Ghid complet de instalare, configurare si operare pentru Cisco si MikroTik

anatolie-ernu/GLPI-LibreNMS

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Contents

1	Despre acest document	2
2	1. Componente	2
2.1	1.1 Ownership-ul datelor	3
3	2. Arhitectura	3
4	3. Cerinte recomandate	3
5	4. Cerinte firewall si routing	4
6	5. Structura repository	4
7	6. Instalarea hostului si Docker	4
7.1	6.1 Pregatirea Ubuntu	4
7.2	6.2 Clonarea repository-ului	5
7.3	6.3 Instalarea Docker	5
8	7. Configurarea .env	5
9	8. Imagini si servicii Docker	5
10	9. Validare si deployment	6
10.1	9.1 Validare	6
10.2	9.2 Deployment	6
11	10. Configurarea initiala GLPI	6
12	11. GLPI Agent	6
12.1	11.1 Pilot	7
12.2	11.2 Rollout GPO	7
13	12. Configurarea initiala LibreNMS	7
14	13. Cisco IOS / IOS XE - SNMPv3	7
15	14. MikroTik RouterOS - SNMPv3	8
15.1	14.1 Enhanced VLAN discovery	9
16	15. Ordinea recomandata de onboarding	9
17	16. Corelarea PC -> IP -> MAC -> VLAN -> Switch -> Port	9
18	17. Securitate si hardening	10

19 18. Reverse proxy si HTTPS	10
20 19. Backup	10
21 20. Restore	11
22 21. Upgrade si mentenanta	11
23 22. Troubleshooting	12
23.1 22.1 Status si logs	12
23.2 22.2 SNMPv3 nu functioneaza	12
23.3 22.3 MAC apare pe uplink	12
23.4 22.4 Duplicate GLPI computers	12
23.5 22.5 Disk usage / restart loop	13
24 23. Runbook operational	13
25 24. Checklist productie	13
26 Anexa A - .env.example	13
27 Anexa B - Configuratie Cisco	14
28 Anexa C - Configuratie MikroTik	14
29 Anexa D - Comenzi utile	15
30 Anexa E - Referinte oficiale	15
31 Anexa F - Glosar	15

1 Despre acest document

Acest ghid reuneste si extinde documentatia repository-ului **GLPI-LibreNMS** intr-un document operational unic. Scopul este implementarea unei platforme self-hosted care poate raspunde la doua intrebari diferite:

1. **Ce este acest asset?** - GLPI.
2. **Unde este conectat in retea?** - LibreNMS.

Rezultatul urmarit este corelarea:

Computer -> IP -> MAC -> VLAN -> Switch -> Physical Port

Stack-ul este destinat in principal retelelor enterprise cu switch-uri **Cisco IOS/IOS XE** si **MikroTik RouterOS**.

Important: un MAC invatat pe un uplink/trunk nu demonstreaza ca endpoint-ul este conectat fizic pe acel port. Pentru localizare se folosesc impreuna FDB/MAC, VLAN, LLDP/CDP si topologia downstream.

2 1. Componente

Componenta	Rol
GLPI	Asset Management / CMDB / inventar hardware si software
GLPI Agent	Inventarierea endpoint-urilor Windows/Linux
LibreNMS	Network discovery, SNMP polling, FDB/MAC, VLAN, LLDP/CDP
MySQL	Baza de date GLPI
MariaDB	Baza de date LibreNMS
Redis	Cache si sesiuni LibreNMS
LibreNMS Dispatcher	Polling/discovery sidecar

2.1 1.1 Ownership-ul datelor

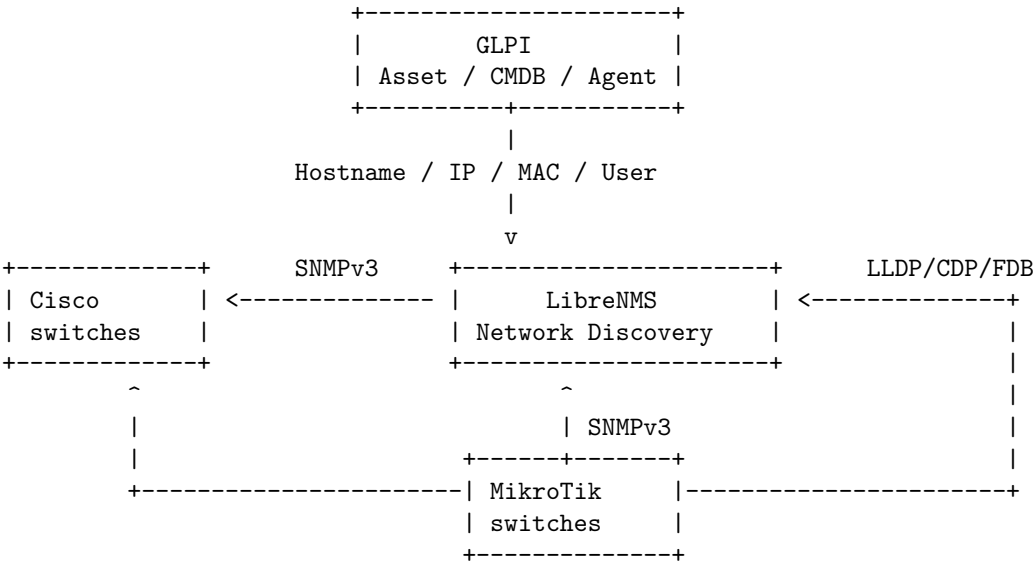
GLPI este autoritativ pentru:

- calculatoare si servere;
- serial number, vendor, model;
- CPU, RAM, storage;
- OS si software instalat;
- user/departament/locatie;
- lifecycle si asset metadata;
- IP/MAC raportate de GLPI Agent.

LibreNMS este autoritativ pentru:

- starea curenta a switch-urilor/routerelor;
- interfaces si counters;
- VLAN-uri;
- FDB/MAC tables;
- ARP/NDP unde sunt disponibile;
- LLDP/CDP neighbors;
- port utilization/errors;
- device availability si sensors.

3 2. Arhitectura



Docker foloseste retele separate `glpi-net` si `librenms-net`. In Stage 1 nu este necesara comunicarea directa GLPI-LibreNMS. O integrare API poate fi adaugata ulterior ca serviciu separat.

4 3. Cerinte recomandate

Resursa	Baseline recomandat
OS	Ubuntu Server 24.04 LTS
CPU	4-8 vCPU
RAM	16 GB
Storage	200-250 GB SSD minim
Network	IP static in management LAN
DNS/NTP	Configurate si functionale
Backup	Storage separat de host

LibreNMS poate genera o cantitate semnificativa de RRD/history data. Monitorizeaza permanent capacitatea filesystem-ului Docker.

5 4. Cerinte firewall si routing

Flux	Port	Directie	Status
SNMP polling	UDP/161	LibreNMS -> device	Obligatori
ICMP	ICMP	LibreNMS -> device	Recomandat
GLPI web	TCP/8081 implicit	Admin/Agent -> host	Stage 1
LibreNMS web	TCP/8000 implicit	Admin -> host	Stage 1
SNMP traps	UDP/162	Device -> LibreNMS	Optional
Syslog	UDP/514	Device -> LibreNMS	Optional
HTTPS	TCP/443	User/Agent -> reverse proxy	Productie

Recomandari:

- SNMP trebuie permis numai din IP-ul LibreNMS.
- MySQL, MariaDB si Redis nu se publica in exterior.
- Porturile 8081/8000 nu se expun pe Internet.
- Foloseste un VLAN de management si routing stabil catre echipamente.

6 5. Structura repository

```
GLPI-LibreNMS/  
  README.md  
  SECURITY.md  
  LICENSE  
  compose.yml  
  .env.example  
  docs/  
    INSTALL.md  
    ARCHITECTURE.md  
    CISCO-SNMPV3.md  
    MIKROTIK-SNMPV3.md  
    GLPI-AGENT.md  
    BACKUP-RESTORE.md  
    TROUBLESHOOTING.md  
    COMPLETE-GUIDE-RO.md  
  config/  
    cisco/snmpv3-example.txt  
    mikrotik/snmpv3-example.rsc  
  scripts/  
    install-docker.sh  
    validate.sh  
    deploy.sh  
    status.sh  
    backup.sh  
  .github/workflows/
```

7 6. Instalarea hostului si Docker

7.1 6.1 Pregatirea Ubuntu

```
sudo apt update  
sudo apt full-upgrade -y
```

```
sudo reboot
```

Configureaza IP static, DNS, NTP si hostname/FQDN inainte de deployment.

7.2 6.2 Clonarea repository-ului

```
cd /opt
sudo git clone https://github.com/anatolie-ernu/GLPI-LibreNMS.git
sudo chown -R $USER:$USER /opt/GLPI-LibreNMS
cd /opt/GLPI-LibreNMS
```

7.3 6.3 Instalarea Docker

```
sudo bash scripts/install-docker.sh
```

Verificare:

```
docker --version
docker compose version
systemctl status docker --no-pager
```

Daca utilizatorul a fost adaugat in grupul **docker**, executa logout/login inainte de rularea Docker fara **sudo**.

8 7. Configurarea .env

```
cd /opt/GLPI-LibreNMS
cp .env.example .env
chmod 600 .env
nano .env
```

Schimba obligatoriu:

```
GLPI_DB_PASSWORD
LIBRENMS_DB_PASSWORD
PUID
PGID
```

Genereaza parole puternice:

```
openssl rand -base64 32
```

Verifica UID/GID:

```
id
```

Nu comite niciodata **.env** in Git. Repository-ul include **.gitignore** si verificare CI, dar acestea nu inlocuiesc secret management-ul operational.

9 8. Imagini si servicii Docker

Configuratia curenta din **.env.example** foloseste:

Serviciu	Imagine
GLPI	glpi/glpi:11.0.8
GLPI DB	mysql:8.4
LibreNMS	librenms/librenms:latest
LibreNMS DB	mariadb:10
Redis	redis:7.2-alpine

Volume persistente:

- `glpi_db` - MySQL GLPI;
- `glpi_data` - `/var/glpi`;
- `librenms_db` - MariaDB LibreNMS;
- `librenms_data` - `/data`, inclusiv `RRD/config/history`.

Pentru productie controlata este recomandat sa testezi toate upgrade-urile in staging. Poti pin-ui imaginile la tag sau digest dupa validare.

10 9. Validare si deployment

10.1 9.1 Validare

```
bash scripts/validate.sh
```

Scriptul verifica:

- existenta `.env`;
- eliminarea placeholder-elor `CHANGE_ME`;
- `docker compose config`;
- sintaxa Bash a scripturilor.

10.2 9.2 Deployment

```
bash scripts/deploy.sh
```

Verificare:

```
docker compose ps
bash scripts/status.sh
```

Loguri:

```
docker compose logs --tail=200 glpi
docker compose logs --tail=200 glpi-db
docker compose logs --tail=200 librenms
docker compose logs --tail=200 librenms-dispatcher
docker compose logs --tail=200 librenms-db
```

Acces initial:

- GLPI: `http://SERVER-IP:8081`
- LibreNMS: `http://SERVER-IP:8000`

11 10. Configurarea initiala GLPI

1. Acceseaza GLPI si finalizeaza wizard-ul initial, daca apare.
2. Configureaza contul administrativ si elimina credentialele implicite.
3. Configureaza entitati, locatii si naming conventions.
4. Verifica automatic actions si cron worker-ul.
5. Configureaza GLPI Agent.
6. Configureaza LDAP/AD doar dupa validarea inventarului de baza.
7. Activeaza HTTPS inainte de rollout larg al agentilor.

GLPI trebuie folosit pentru asset inventory/CMDB, nu drept inlocuitor pentru FDB/LLDP polling in timp real.

12 11. GLPI Agent

Endpoint implicit pentru GLPI 10+ / 11:

```
http://<GLPI_SERVER>:8081/front/inventory.php
```

Dupa reverse proxy foloseste URL-ul HTTPS canonic.

12.1 11.1 Pilot

1. Instaleaza GLPI Agent pe 2-5 statii reprezentative.
2. Configureaza target-ul catre GLPI.
3. Forteaza inventory.
4. Confirma ca fiecare device apare o singura data.
5. Verifica serial/model/OS/software/IP/MAC.
6. Cauta acelasi MAC in LibreNMS.
7. Valideaza switch/port/VLAN.

12.2 11.2 Rollout GPO

Model recomandat:

Computer Configuration

- > Policies
 - > Software Settings / Startup Script
 - > GLPI Agent MSI installation

Exemplu share:

\\domain.example\NETLOGON\software\glpi-agent\

Rollout rings:

Ring	Populatie
0	IT test devices
1	Departamentul IT
2	Business users selectati
3	Toate statiile administrate
4	Servere, dupa validare separata

Deployment-ul trebuie sa fie idempotent. Nu incorporeaza credentiale administrative in startup scripts.

13 12. Configurarea initiala LibreNMS

1. Acceseaza `http://SERVER-IP:8000`.
2. Creeaza utilizatorul administrator.
3. Ruleaza validarea:

```
docker compose exec librenms lnms validate
```

4. Adauga un singur Cisco pilot.
5. Verifica interfaces, VLAN, FDB/MAC, LLDP/CDP, sensors si graphs.
6. Adauga un MikroTik pilot.
7. Extinde onboarding-ul numai dupa validarea pilotului.

Un device poate fi adaugat si din CLI:

```
docker compose exec librenms lnms device:add <HOSTNAME_OR_IP>
```

14 13. Cisco IOS / IOS XE - SNMPv3

Standard: **SNMPv3 authPriv**, read-only, ACL catre IP-ul LibreNMS.

```
conf t
```

```
ip access-list standard ACL-LIBRENMS-SNMP
permit host <LIBRENMS_IP>
deny any log
```

```
exit
```

```
snmp-server view LIBRENMS-VIEW iso included
snmp-server group LIBRENMS-GROUP v3 priv read LIBRENMS-VIEW access ACL-LIBRENMS-SNMP
snmp-server user librenms LIBRENMS-GROUP v3 auth sha <AUTH_PASSWORD> priv aes 128 <PRIV_PASSWORD>
```

```
snmp-server contact <CONTACT>
snmp-server location <LOCATION>
```

```
lldp run
cdp run
```

```
end
write memory
```

Foloseste parole diferite pentru autentificare si privacy.

Daca setezi manual SNMP engine-id, configureaza-l inainte de crearea userilor SNMPv3.

Test de pe hostul LibreNMS:

```
sudo apt install -y snmp
snmpwalk -v3 \
  -l authPriv \
  -u librenms \
  -a SHA \
  -A '<AUTH_PASSWORD>' \
  -x AES \
  -X '<PRIV_PASSWORD>' \
  <SWITCH_IP> \
  1.3.6.1.2.1.1
```

Verificari Cisco:

```
show snmp user
show snmp group
show access-lists ACL-LIBRENMS-SNMP
show lldp neighbors detail
show cdp neighbors detail
show mac address-table
```

Dupa polling confirma:

- ports corecte;
- VLAN-uri;
- FDB/MAC;
- LLDP/CDP neighbors;
- diferentierea uplink vs edge/access.

15 14. MikroTik RouterOS - SNMPv3

Configuratia de baza este read-only, cu acces limitat la LibreNMS /32.

```
/snmp community
set [find default=yes] read-access=no write-access=no
add name="librenms" \
  addresses=<LIBRENMS_IP>/32 \
  security=private \
  authentication-protocol=SHA1 \
  authentication-password="<AUTH_PASSWORD>" \
  encryption-protocol=AES \
```



```

encryption-password="<PRIV_PASSWORD>" \
read-access=yes \
write-access=no

```

```

/snmpp
set enabled=yes contact="<CONTACT>" location="<LOCATION>"

```

Optional, daca raspunsul pleaca cu source address gresit:

```

/snmpp set src-address=<MANAGEMENT_IP>

```

Verificari:

```

/snmpp print detail
/snmpp community print detail
/interface bridge host print
/ip neighbor print detail

```

Test SNMP din hostul LibreNMS este identic cu Cisco, folosind IP-ul MikroTik.

15.1 14.1 Enhanced VLAN discovery

LibreNMS poate utiliza helper-ul RouterOS LNMS_vlans pentru detalii VLAN. Acesta poate necesita SNMP write pentru a declansa un script RouterOS.

Repository-ul lasa write-access dezactivat implicit. Daca functia este necesara:

1. Restrictioneaza sursa la LibreNMS /32.
2. Pastreaza SNMPv3 authPriv.
3. Review-ueste scriptul.
4. Oferă numai policy-urile necesare.
5. Documenteaza exceptia ca acces privilegiat.
6. Monitorizeaza configuration changes.

16 15. Ordinea recomandata de onboarding

1. Un Cisco access switch.
2. Validare SNMPv3, interfaces, VLAN, FDB/MAC, LLDP/CDP.
3. Un MikroTik.
4. Validare SNMPv3, bridge/FDB si neighbors.
5. Core/distribution switches.
6. Restul access switches.
7. GLPI Agent pe cateva endpoint-uri.
8. Corelare end-to-end pentru 5-10 endpoint-uri.
9. Rollout larg al GLPI Agent.
10. Optional traps/syslog dupa stabilizarea polling-ului.

17 16. Corelarea PC -> IP -> MAC -> VLAN -> Switch -> Port

Exemplu:

GLPI

```

Hostname: HQ-PC-043
IP:      10.129.21.43
MAC:     00:25:90:AB:CD:EF

```

LibreNMS

```

cauta MAC 00:25:90:AB:CD:EF
|
+-- access port -> endpoint probabil gasit

```

```

|
+-- uplink/trunk -> urmeaza LLDP/CDP catre downstream switch
|
+-- repeta FDB lookup

```

Rezultat:

```

Switch: HQ-ACCESS-SW-04
Port:   Gi1/0/31
VLAN:   129

```

Reguli:

- confirma ca FDB entry este recenta;
- confirma VLAN si port state;
- daca portul este trunk, continua downstream;
- coreleaza GLPI Agent IP/MAC cu ARP/FDB;
- tine cont de AP-uri, telefoane IP, mini-switch-uri si bridge-uri.

18 17. Securitate si hardening

- SNMPv3 `authPriv` pentru Cisco si MikroTik.
- SNMP read-only implicit.
- ACL/restrictie /32 la LibreNMS.
- `.env` cu `chmod 600`, niciodata in Git.
- Databases si Redis fara porturi publice.
- GLPI si LibreNMS in management LAN.
- HTTPS pentru productie.
- MikroTik SNMP write dezactivat implicit.
- Backup separat si restore tests.
- Upgrade numai dupa staging validation.

Nu folosi `community public` sau `SNMPv2c` in productie daca echipamentul suporta `SNMPv3`.

19 18. Reverse proxy si HTTPS

In productie foloseste FQDN-uri interne, de exemplu:

Serviciu	FQDN	Backend
GLPI	<code>glpi.example.local</code>	<code>http://docker-host:8081</code>
LibreNMS	<code>librenms.example.local</code>	<code>http://docker-host:8000</code>

Recomandari:

- certificat emis de CA interna sau ACME;
- preserve `Host` si `X-Forwarded-*` headers;
- evita SSL inspection care rupe certificatul pentru GLPI Agent;
- activeaza HSTS numai dupa validarea completa HTTPS;
- actualizeaza target-ul GLPI Agent dupa migrare.

20 19. Backup

`bash scripts/backup.sh`

Backup-ul include:

- GLPI database dump;
- LibreNMS database dump;

- /var/glpi persistent data;
- /data LibreNMS persistent data;
- compose.yml;
- template .env;
- SHA256SUMS.

Fisierul real .env nu este copiat intentionat.

Verificare:

```
cd <BACKUP_DIRECTORY>
sha256sum -c SHA256SUMS
```

21 20. Restore

Ordine:

1. Opreste serviciile aplicative.
2. Restaureaza datele persistente.
3. Restaureaza bazele de date.
4. Porneste stack-ul.
5. Ruleaza `lnms validate`.
6. Verifica un endpoint GLPI si un switch.

Exemplu:

```
BACKUP=/opt/backups/glpi-librenms/20260914_150000
```

```
docker compose stop glpi librenms librenms-dispatcher
```

```
gzip -dc "$BACKUP/glpi-data.tar.gz" | \
  docker compose run --rm -T glpi sh -lc 'rm -rf /var/glpi/* && tar -C /var/glpi -xzf -'
```

```
gzip -dc "$BACKUP/librenms-data.tar.gz" | \
  docker compose run --rm -T librenms sh -lc 'rm -rf /data/* && tar -C /data -xzf -'
```

```
gzip -dc "$BACKUP/glpi-db.sql.gz" | \
  docker compose exec -T glpi-db sh -lc \
  'mysql -u"$MYSQL_USER" -p"$MYSQL_PASSWORD" "$MYSQL_DATABASE"'
```

```
gzip -dc "$BACKUP/librenms-db.sql.gz" | \
  docker compose exec -T librenms-db sh -lc \
  'mariadb -u"$MYSQL_USER" -p"$MYSQL_PASSWORD" "$MYSQL_DATABASE"'
```

```
docker compose up -d
```

```
bash scripts/status.sh
```

```
docker compose exec librenms lnms validate
```

Un backup care nu a fost restaurat niciodata nu este demonstrat. Executa periodic restore tests si inregistreaza RPO/RTO.

22 21. Upgrade si mentenanta

Flux recomandat:

1. Citeste release notes.
2. Executa backup.
3. Verifica checksums.
4. Testeaza in staging.
5. Actualizeaza tag/digest numai dupa validare.

6. Ruleaza pull/up.
7. Verifica status si `lnms validate`.
8. Testeaza login, inventory si polling.
9. Pastreaza rollback plan.

```
bash scripts/backup.sh
docker compose pull
docker compose up -d
bash scripts/status.sh
docker compose exec librenms lnms validate
```

23 22. Troubleshooting

23.1 22.1 Status si logs

```
docker compose ps
bash scripts/status.sh

docker compose logs --tail=200 glpi
docker compose logs --tail=200 glpi-db
docker compose logs --tail=200 librenms
docker compose logs --tail=200 librenms-dispatcher
docker compose logs --tail=200 librenms-db
```

23.2 22.2 SNMPv3 nu functioneaza

Verifica:

- routing catre management IP;
- UDP/161;
- username;
- authPriv;
- auth algorithm/password;
- privacy algorithm/password;
- ACL Cisco sau MikroTik /32;
- source IP real al LibreNMS;
- VRF/routing table folosit de device.

```
snmpwalk -v3 \
-l authPriv \
-u librenms \
-a SHA \
-A '<AUTH_PASSWORD>' \
-x AES \
-X '<PRIV_PASSWORD>' \
<DEVICE_IP> \
1.3.6.1.2.1.1
```

23.3 22.3 MAC apare pe uplink

Este normal. Urmareste LLDP/CDP/neighbor topology downstream si repeta FDB lookup pana la portul edge final.

23.4 22.4 Duplicate GLPI computers

Verifica:

- agent identity;
- serial number quality;
- Windows cloned images;
- assets create manual;

- reinstall/enrollment history.

Nu sterge duplicatele pana nu stabilesti recordul autoritativ.

23.5 22.5 Disk usage / restart loop

```
df -h
docker system df
docker volume ls
docker inspect <container>
docker compose logs --tail=300 <service>
```

24 23. Runbook operational

Frecventa	Actiune
Zilnic	Alerte LibreNMS, restart/unhealthy, disk critic
Saptamanal	Device down, polling errors, GLPI agent failures, backup status
Lunar	Patching host, image review, capacity trend, access review
Trimestrial	Restore test, SNMP ACL/users, certificate expiry
Schimbare majora	Backup, staging, change ticket, rollback plan

Quick health check:

```
cd /opt/GLPI-LibreNMS
bash scripts/status.sh
docker compose exec librenms lnms validate
df -h
docker system df
```

25 24. Checklist productie

- ☐ Ubuntu 24.04 patch-uit, IP static, DNS si NTP OK.
- ☐ Docker Engine si Compose plugin instalate.
- ☐ .env cu parole puternice si chmod 600.
- ☐ GLPI si LibreNMS fara restart loop.
- ☐ GLPI functional si cron worker activ.
- ☐ LibreNMS lnms validate fara FAIL critic.
- ☐ Cisco pilot SNMPv3 authPriv + ACL + LLDP/CDP validat.
- ☐ MikroTik pilot SNMPv3 + /32 + write disabled validat.
- ☐ FDB/MAC si VLAN validate.
- ☐ GLPI Agent pilot pe 2-5 endpoint-uri.
- ☐ Corelare GLPI MAC -> LibreNMS switch/port confirmata.
- ☐ Reverse proxy + HTTPS configurat.
- ☐ Backup automatizat.
- ☐ Restore test executat.
- ☐ DB/Redis nu sunt expuse extern.
- ☐ CI repository green.

26 Anexa A - .env.example

```
# General
TZ=Europe/Chisinau
PUID=1000
```

```

PGID=1000

# Published ports
GLPI_HTTP_PORT=8081
LIBRENMS_HTTP_PORT=8000

# Images
GLPI_IMAGE=glpi/glpi:11.0.8
GLPI_DB_IMAGE=mysql:8.4
LIBRENMS_IMAGE=librenms/librenms:latest
LIBRENMS_DB_IMAGE=mariadb:10
REDIS_IMAGE=redis:7.2-alpine

# GLPI database
GLPI_DB_NAME=glpi
GLPI_DB_USER=glpi
GLPI_DB_PASSWORD=CHANGE_ME_STRONG_GLPI_DB_PASSWORD
GLPI_CRONTAB_ENABLED=1
GLPI_SKIP_AUTOINSTALL=false
GLPI_SKIP_AUTOUPDATE=false

# LibreNMS database
LIBRENMS_DB_NAME=librenms
LIBRENMS_DB_USER=librenms
LIBRENMS_DB_PASSWORD=CHANGE_ME_STRONG_LIBRENMS_DB_PASSWORD

# LibreNMS PHP/runtime settings
LIBRENMS_MEMORY_LIMIT=512M
LIBRENMS_MAX_INPUT_VARS=2000
LIBRENMS_UPLOAD_MAX_SIZE=32M
LIBRENMS_OPCACHE_MEM_SIZE=256

# Deployment metadata
BACKUP_DIR=/opt/backups/glpi-librenms

```

27 Anexa B - Configuratie Cisco

Fisier repository: config/cisco/snmpv3-example.txt.

```

conf t
ip access-list standard ACL-LIBRENMS-SNMP
  permit host <LIBRENMS_IP>
  deny any log
exit
snmp-server view LIBRENMS-VIEW iso included
snmp-server group LIBRENMS-GROUP v3 priv read LIBRENMS-VIEW access ACL-LIBRENMS-SNMP
snmp-server user librenms LIBRENMS-GROUP v3 auth sha <AUTH_PASSWORD> priv aes 128 <PRIV_PASSWORD>
snmp-server contact <CONTACT>
snmp-server location <LOCATION>
lldp run
cdp run
end
write memory

```

28 Anexa C - Configuratie MikroTik

Fisier repository: config/mikrotik/snmpv3-example.rsc.

```

/snmpp community
set [find default=yes] read-access=no write-access=no
add name="librenms" addresses=<LIBRENMS_IP>/32 security=private authentication-protocol=SHA1 authentication
/snmpp
set enabled=yes contact="<CONTACT>" location="<LOCATION>"
# /snmp set src-address=<MANAGEMENT_IP>
/snmpp print detail
/snmpp community print detail

```

29 Anexa D - Comenzi utile

Comanda	Scop
sudo bash scripts/install-docker.sh	Instaleaza Docker
bash scripts/validate.sh	Valideaza configuratia
bash scripts/deploy.sh	Deploy stack
bash scripts/status.sh	Status + HTTP checks
bash scripts/backup.sh	Backup complet
docker compose exec librenms lnms validate	Validate LibreNMS
docker compose logs --tail=200 <service>	Troubleshooting

30 Anexa E - Referinte oficiale

- Repository: <https://github.com/anatolie-ernu/GLPI-LibreNMS>
- GLPI Docker: <https://github.com/glpi-project/docker-images>
- GLPI documentation: <https://help.glpi-project.org/documentation/>
- GLPI Agent: <https://glpi-agent.readthedocs.io/>
- LibreNMS Docker: <https://docs.librenms.org/Installation/Docker/>
- LibreNMS documentation: <https://docs.librenms.org/>
- Docker Engine Ubuntu: <https://docs.docker.com/engine/install/ubuntu/>
- MikroTik RouterOS documentation: <https://manual.mikrotik.com/>
- Cisco documentation: <https://www.cisco.com/>

31 Anexa F - Glosar

Termen	Explicatie
CMDB	Configuration Management Database
FDB	Forwarding Database / tabela MAC
ARP	Mapare IPv4 -> MAC
LLDP	Link Layer Discovery Protocol
CDP	Cisco Discovery Protocol
SNMPv3 authPriv	SNMP cu autentificare si criptare
RRD	Round Robin Database
Access port	Port edge asociat endpoint-ului
Trunk/uplink	Port intre switch-uri / multiple VLAN-uri
RPO	Recovery Point Objective
RTO	Recovery Time Objective